

GIS-based geocoding methods for area-based addresses and 3D addresses in urban areas

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Abstract. For more than four decades, two address-matching methods, the street-based address geocoding method and address-point-matching method, have been used to identify geographical coordinates from postal addresses. However, street-based address geocoding methods developed for the US addressing system are not universally applicable in developing a single-portal geocoding middleware for worldwide Internet-based geographic information systems applications. Problems also exist with address-point matching especially in its capability to identify features and incorporate 3D locational data from 3D addresses for analysis of large public buildings, shopping centers or metro-subways, that exist in urban environments. To alleviate these problems, this paper details two alternative address-matching methods, an area-based address geocoding method and a 3D address geocoding method. The area-based address geocoding method is a 2D positioning method based on a 2D area-based address-matching technique. The 3D address geocoding method is based on a universally applicable 3D address-geocoding technique. To elaborate, this paper introduces (1) a BlockObject model and designs reference databases for an area-based address system, (2) a 3D indoor network model representing the internal structures of urban environment, and (3) a 3D address geocoding algorithm based on a 3D indoor geocoding method. To illustrate the benefits of the 3D address-positioning method, this paper implements 3D indoor navigation to define optimal routes within a single building.

1 Introduction

Postal addresses are one of several sources of geospatial information that may be used to create administrative or business databases that contain individual-level activity data. To identify the geographic coordinates on the databases, the geographic information systems (GIS) technology has relied on an address-matching method, also known as geocoding (Drummond, 1995). It can convert administrative or business street-address databases, collected and stored by governments, utilities, and the private sector, into a GIS database containing locational information. Geocoding postal addresses allows social scientists to obtain disaggregated data and permits analysis of a broad array of individual human activity and movement patterns. A wide range of GIS applications and research areas includes studies of travel behavior, migration and residential mobility, pedestrian behavior choice, as well as shopping and commuting behavior (Kwan, 1998; 2000; Kwan and Lee, 2004; Miller, 1999; Weber and Kwan, 2002).

Address-matching techniques were developed initially by the US Census Bureau for the specific purpose of allocating population accurately within blocks, census tracts, and other geographic areas, without the expense of sending enumerators to every dwelling unit in the country (Drummond, 1995). The street-based address-matching method, however, was developed for use in the US. Although the street-addressing system has been used in most Western countries, an area-based addressing system, and a hybrid addressing system are used in other parts of the world (Atkinson, 1999). Hierarchical area-based addressing systems are used in Eastern Asia (Davis et al, 2003), while hybrid addressing systems are used in China, and India uses a street-addressing

system containing suburban or district names. Because no established geocoding methods are available for the area-based and hybrid addressing systems, some counties have developed address-rationalization programs to reassign their addressing schema based on the street-based addressing system (MOGAHA, 2003). The street-based address geocoding method developed for the US is exclusive to certain countries and therefore, it has limitations in developing worldwide Internet-based GIS applications such as GoogleEarth. Such applications require the unified portal of a geocoding middleware with a common user interface (polymorphism in object orientation) to identify geographic positions from any given type of worldwide address data.

In order to resolve this problem, an address-point-matching approach has been developed in the United Kingdom by the Ordnance Survey (2004; 2006). This approach requires and uses a point-georeferenced individual-address layer as the reference-address database. The matching process establishes corresponding relationships between input postal addresses and all geographic information in the reference database. An individual point in the reference database can represent a single-building postal address, which can be described by any addressing system, including either street-based or area-based addressing systems. Despite these widespread advantages, this matching method cannot be used for 3D addressing. This limitation occurs because 3D addressing represents multiple apartment or suite addresses for a building, as a subunit number of the building address. The address-point-matching method cannot identify 3D locational data from 3D addresses for analyzing human activities in urban environments such as large public buildings, shopping centers or metro subways (Kwan and Lee, 2005; Pu and Zlatanova, 2005; Winter et al, 2005).

In order to alleviate the problems, this paper proposes two address-matching methods, which are called an area-based address geocoding method and a 3D address geocoding method. The area-based address geocoding method is a 2D positioning method based on a 2D area-based address-matching technique and the 3D address geocoding method is based on a universally applicable 3D address geocoding technique. In order to demonstrate the benefits of using the 3D address-positioning method, this paper implements 3D indoor navigation to define optimal routes within microscale urban environments, as a useful by-product of the system.

The paper is organized as follows. Section 2 reviews possibilities and limitations in current addressing systems and geocoding methods. Section 3 proposes a new 2D area-based address-matching method and a 3D address-positioning method by utilizing conventional address-matching technology (Corbett, 1979) and a 3D indoor geocoding method (Lee, 2004b). The fourth section describes the output from the experimental implementation of the 3D address-matching technique and demonstrates the potential benefits of this technique and 3D indoor navigation for providing location-based services. The final section discusses several significant substantive insights derived from this study.

2 Current geocoding methods and processes

Geocoding is defined as the process of converting addresses, place names, or other geographic references to geographic coordinates, such as latitude and longitude (Goodchild, forthcoming). One of the most popular geocoding methods is address matching in GIS.

Although addressing systems have been recognized as a research problem, consistency in standardizing and formatting address data varies considerably from one country to another (Atkinson, 1999). The first addressing initiatives took place in Western Europe and China in the 18th century (Davis et al, 2003). While subsequent addressing systems were employed throughout the world, few standardized addressing systems

have been developed. According to the Universal Postal Union, a specialized agency of the United Nations developing a POST*CODE[®] product to search for postal addressing systems in 190 member countries (UPU, 2004), each country has slightly different address formats. The address formats can be classified into three systems: a street-network-based address system, an area-based address system, and a hybrid address system.

The geocoding process includes three steps: parsing, matching, and locating steps. Parsing deals with the semistructured input of addresses of events, matching creates a correspondence between the input address and the address-reference database, and finally, locating assigns coordinates of the event, as seen in figure 1. In order to perform the three steps, the geocoding process needs a reference database in which information about the addressing system is stored, so that geocoding methods can be performed in several ways across the available address-reference databases.

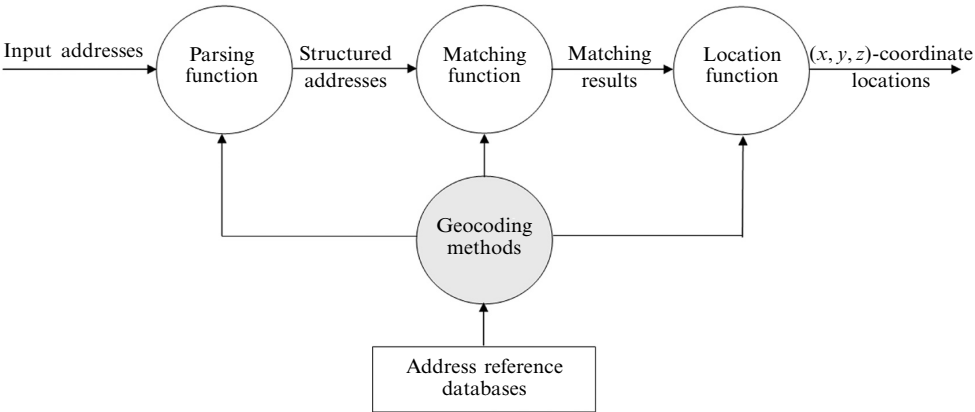


Figure 1. Geocoding dataflow diagram.

The parsing step converts the addresses in an input database into a structured and standardized format ready for the next step. Input databases generally store address information in nonstandardized formats, and address data within the same database will be formatted inconsistently. In the address field of an input database, characteristic problems include abbreviated street names, misspelled street names, complex directional combinations, and confusing or ambiguous information. The parsing step can be executed through various text-orientated and string-oriented algorithms (Davis et al, 2003) using translation tables and the Soundex function (Knuth, 1973).

The matching step tries to find the most precise reference information from the reference-address database, which is associated with a given input address. Two types of reference-address databases such as street centerlines with address ranges (US Census, 1992) and point-georeferenced individual addresses (Ordnance Survey, 2004) have been used. When using street centerlines with address ranges as the reference database, that is, the TIGER/Line files developed by the US Census Bureau, the matching step identifies a street-segment record in the reference database that has the same street name, street type, and other identifiers as the record in the input address. After the two records have been matched, interpolation is used to assign geographic coordinates to the input address in the locating step (Walls, 2003).

In the United Kingdom the Ordnance Survey produces an address-point database, called the address layer, that provides precise coordinates for more than twenty-six million residential and commercial properties in the United Kingdom, on the basis of addresses recorded in the Royal Mail’s postcode address file (PAF[®])—the most

accurate and up-to-date link between any property address and its location on the map (Ordnance Survey, 2004). In general, this step is the most precise method for matching the input address to an object from the point-georeferenced individual addresses.

Finally, the locating step receives the results from the matching step, and determines the actual geographic coordinates that are to be assigned to the input address. Depending upon the reference-address database, the locating step either copies the coordinates of a point object in the individual point address database, or interpolates to determine the address's approximate distance along the street segment by using the low and high addresses of the block face as the well-known street-address-matching technique (US Census, 1992).

As mentioned in the previous section, geocoding methods should be developed and implemented on the basis of their addressing systems. The street-based address geocoding method has been developed for the US addressing system, and the address-point-matching method has a limitation in identifying 3D locational data from 3D postal addresses. These geocoding methods are not suitable for either area-based addresses or 3D addresses. Therefore, this paper proposes to develop a new 2D area-based address-matching method and 3D address geocoding method, as well as designing address databases to serve as reference datasets in the geocoding process.

3 Geocoding methods for area-based addresses and 3D addresses

Two important procedures are required to identify the locational information of address-based data using address-matching methods. One is to model urban environments to serve as reference data for geocoding address-based data. The other is to implement a geocoding method to provide the location of the input data. On the basis of the object-orientation method, object–object relationships of the addressing model are represented using a unified modeling language (UML) object diagram. Figure 2 shows the classes and their relationships to the model. An *AddressObject* class has three subclasses: *AreaAddressObject*, *StreetAddressObject*, and *PointAddressObject*. The inheritance relationship between the *AddressObject* and subclasses is indicated by an arrow with a solid arrowhead pointing to the parent class. The *AreaAddressObject* describes the locational information of geographic objects on the basis of the hierarchical relationships of areas, called a hierarchical area-based addressing system. The *PointAddressObject* is a single building that has postal addresses represented in a point-georeferenced individual-address database.

The *MetropolisObject* and *PostalCodeObject* classes are associated with the *AreaAddressObject* class through a composition relationship that models the case where the *MetropolisObject* and *PostalCodeObject* are part of the *AreaAddressObject*. The composition relationship is indicated by a solid diamond headed arrow pointing from the part to the whole, and the cardinality of composition is indicated in the figure. Because the area-based addressing system is based on hierarchical areas, the municipalities are subdivided from big to small, into metropolis (*to* or *fu* in Japan), ward (*ku*), district (*cho*), neighborhood (*chome*), and block (*ban*). Due to the hierarchical relations of objects, the object–object relationship between a class and a subclass is an aggregation relationship indicated by a line with a hollow diamond. In other words, the metropolis area is composed of many ward regions, and the ward has many districts, and so on. A *BlockObject* is associated with *NeighborhoodObject* through an aggregation relationship. The *BlockObject* has attributes including *Block_ID*, block number, house number ranges, and attributes inherited from nested parent classes.

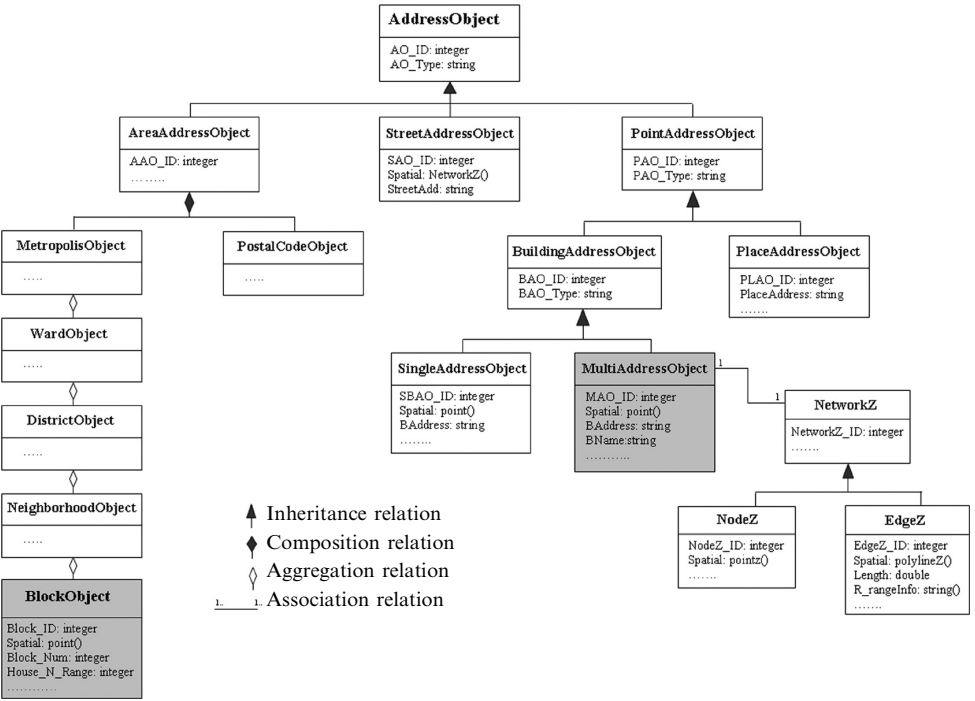


Figure 2. Unified modeling language object diagram of address systems.

The PointAddressObject class has two subclasses: BuildingAddressObject and PlaceAddressObject, indicated by an inheritance relationship in the UML object diagram. The BuildingAddressObject class has two subclasses: SingleAddressObject and MultiAddressObject, which share properties and attributes with the parent class, the BuildingAddressObject. The association relationship between the MultiAddressObject and a NetworkZ class is modeled by a line connecting the two classes in figure 2. The BlockObject and MultiAddressObject classes are defined in the following section.

3.1 2D Geocoding method for an area-based addressing system

In an area-based addressing system, the address is written in the order from the smallest unit to the largest unit. For example, the address of a single house in Tokyo, Japan is 10–23, 5-chome, Minaniazabu, Minato-ku, Tokyo 106-001. The last number (106-001) is the postal code, and 23 is the building (house) number (23-go). 10 is the block number (10-ban) and 5 is the neighborhood number (5-chome). The district name is Minaniazabu and the ward name is Minato-ku. In order to geocode area-based addresses, two steps are proposed: one is to geocode the address on the basis of the areal address (for example, 10-ban, 5-chome, Minaniazabu, Minato-ku, Tokyo) to define the location information of the block, and the next is to define the positioning of the building within the block on the basis of the building number (23). Since the first step of area-based address matching can be implemented by attribute and spatial queries (ESRI, 2001) on the basis of municipality-boundary codes (such as Ward_Code, District_Code, Neighborhood_Code and Block_Code), this paper presents the area-based address geocoding method to define the house location within a block.

Area-based geocoding methods used to identify the location information of the individual objects within a polygon are (1) the generation of a random point inside

the area (block) boundaries (Davis et al, 2003) or (2) an area centroid geocoding method (ESRI, 2001). This paper proposes a new way to geocode building addresses within block boundaries, called the area-based address geocoding method for area-based address systems.

The first procedure is to model the block boundaries to provide a reference database to geocode building numbers within a block. Since a polygon is described using connected lines in the vector data model for geometric representations, the block polygon structure is identical to the polyline structure (ESRI, 1998). A polygon is a connected sequence of three or more points that form a closed, nonintersecting loop. The primary classes of the BlockObject model (BOM) are Point and BlockObject, whose schema is shown in figure 3. The block structure consists of a set of Points pt and other attributes including an identifier, and the total number of points. In addition, two attributes store a first house (or building) number associated with the first point of the set of Points (Point[0]) and a last house number within the block, and this information will be used for the 2D geocoding explained in the following section. Points for the block polygon always appear in a clockwise order and the first and last point of the block must be the same. Figure 3 describes the implementation of the BlockObject model in a relational database model, and depicts the complete logical diagram of the database structure. Each block of the BlockObject class in the database has an identifier (Block_ID), a block name, and four area codes, which are foreign keys of the parent classes: the NeighborhoodObject, the DistrictObject, the WardObject, and MetropolisObject classes. The BlockObject has the building number (or lot number) range information that serves as a reference to identify the geographic positioning of a given address. This procedure is called an area-based geocoding method, which is discussed in the following section.

The fundamental principle of the geocoding method for area-based addresses employs a well-known street-address-matching technology based on a linear interpolation method. The geocoding method has an explicit assumption that all parcels (or buildings) on the same block have the same length of frontage. A TIGER/Line reference database model can be comprised of the BlockObject Model. A block in the model contains building numbers as a range attribute, which refers to a first house number (FHouseNum) on the first point of the block and a last house number

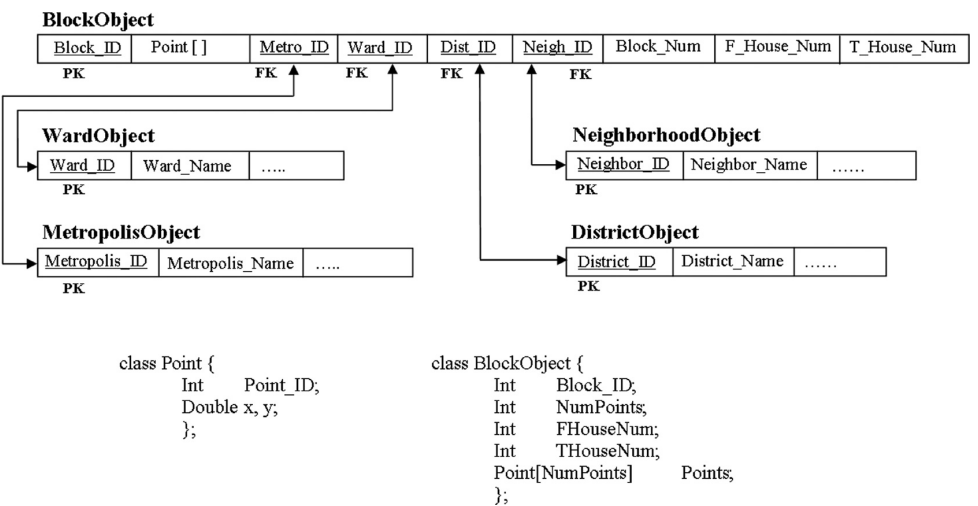


Figure 3. A BlockObject model and relational database schema.

(THouseNum) on the end point of the block. In general, each house within the block (10-ban) is uniquely labeled. The house labels provide a useful value in a database table because the labels are associated with parcel numbers in a Cadastre system (Lee, 2004b). The house number (or parcel number) is not always assigned as a number in sequence, unlike a street address, because the houses may be numbered in construction order (Bullock, 1999). In order to standardize the house numbers (using sequential numbering in clockwise order), translation tables can be used (Drummond, 1995). For example, a translation table would change 23, 24-1, and 24-2 in clockwise order into 23, 24, and 25, respectively, before geocoding the house address within the block. After declining the house location on the basis of the standardized house number, the georeferenced point will contain the original house address (input address) by simply copying the untranslated housing address.

In order to identify the location of a single family-house address, the 2D area-based address geocoding method is executed in seven steps. The entire procedure for geocoding the parsed house address within the block is summarized as follows.

Procedure. 2DArea-basedAddressGeocoding (HouseAdd h_add , BlockObject $blockList$).

Step 1: Split the given house address (h_add) into a house number (k) and a block address (b_add).

Step 2: Based on the block address (b_add), retrieve a block i (B_i) from the set of blocks ($blockList$).

Step 3: Calculate the network distance (d_{0k}) from a starting point ($pt_0 = \text{Point}[0]$) along the boundary of block i (B_i) based on the target house number (k) and the house number ranges of blocks in BOM. In the case where the house numbers are not arranged sequentially in block i , the house address (h_add) needs to be standardized using a translation table.

Step 4: Identify a segment of block i (B_i) according to $d_{0j} < d_{0k} < d_{0j+1}$, where d_{0j} is the network distance from pt_0 to pt_j in block i (B_i).

Step 5: Determine the locational positioning (n_k) of house number k [with (x , y)-coordinates] along the boundary of block i (B_i).

Step 6: Apply a user-specified offset to move the point location (representing a house) a certain distance (inside) away from the boundary of block i (B_i).

Step 7: Determine the locational positioning (n_k) of house number k [with (x , y)-coordinates] with the offset. Check whether the node (n_k) is located inside block i (B_i) using a point-in-polygon query.

To place these steps in an example, suppose that we try to geocode a single house in Tokyo, Japan, with address 10-23, 5-chome, Minaniazabu, Minato-ku, Tokyo 106-001. In the first step, the given address (h_add) is subdivided into a house number ($k = 23$) and the block address (b_add), 10-ban, 5-chome, Minaniazabu, Minato-ku, Tokyo. The next step is to retrieve a block polygon (B_i) by an attribute query based on the block address (b_add). The following steps are to identify the location of house k in the block model (BOM) for a block i . The distance d_{0k} is equal to distance d_{0t} multiplied by the proportion of house-number ranges $[(HouseNum_k - FHouseNum) / (THouseNum - FHouseNum)]$, where t is the total number of points ($NumPoints$) in B_i . The distance d_{0t} is the sum of all the segment lengths in the block polygon B_i . The next step identifies a segment of block i (B_i) according to $d_{0j} \leq d_{0k} \leq d_{0j+1}$, where d_{0j} is a network distance from pt_0 to pt_j in block i (B_i). Suppose $d_{02} < d_{0k} < d_{03}$, the segment s_{23} connecting pt_2 and pt_3 is identified, as well as the distance d_{2k} calculated by $d_{0k} - d_{02}$.

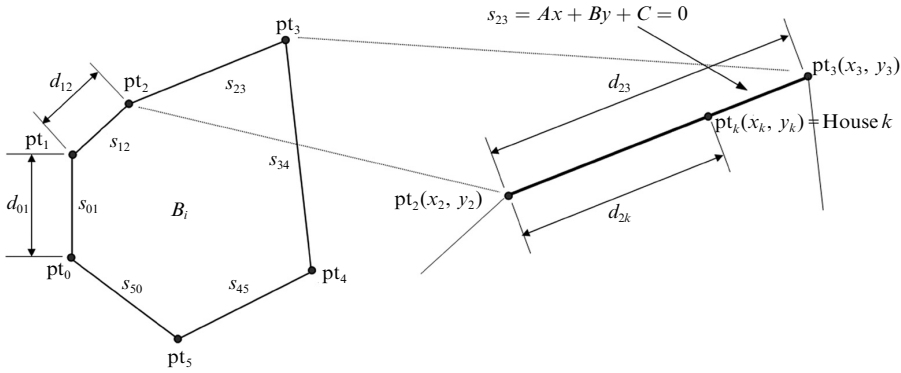


Figure 4. Area-based address geocoding method.

in figure 4. Segment s_{23} connecting node $n_2(x_2, y_2)$ to node $n_3(x_3, y_3)$ is characterized by the direction of a vector format $\mathbf{v} = a_x \mathbf{i} + a_y \mathbf{j}$, where $a_x = x_3 - x_2$, and $a_y = y_3 - y_2$. The (x, y) -coordinates of point pt_k (representing house k) are calculated by the mathematical functions described below. $\mathbf{u}$ is a unit vector of the segment s_{23} .

$$\text{Distance } d_{0i} = \sum_{j=0}^i d_{jj+1} .$$

$$d_{0k} = d_{0t} \times \frac{(\text{HouseNum}_k - \text{FHouseNum})}{(\text{THouseNum} - \text{FHouseNum})}, \text{ where } t = \text{PointNum in block } B_i .$$

If

$$(d_{02} \leq d_{0k} \leq d_{03}),$$

then

$$d_{2k} = d_{0k} - d_{02} .$$

For

$$pt_k(x_k, y_k): x_k = d_{2k} \times (x_u + x_i), y_k = d_{2k} \times (y_u + y_i),$$

where, $\mathbf{u}(x_u, y_u)$ is the unit vector of a segment s_{ij} ,

$$x_u = \frac{a_x}{(a_x^2 + a_y^2)^{1/2}},$$

$$y_u = \frac{a_y}{(a_x^2 + a_y^2)^{1/2}},$$

$$\mathbf{v} = a_x \mathbf{i} + a_y \mathbf{j} = (x_j - x_i) \mathbf{i} + (y_j - y_i) \mathbf{j},$$

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} .$$

The final step is to apply a user-specified offset to move the point location (representing a centroid of house location within the block) a certain distance (inside) away from segment s_{23} (a block boundary). As seen below, a vector method for identifying a right offset of the point pt_k is to calculate the edge-vector cross product (Lee, 2004a), in order to define the point location inside the block. Vector $\mathbf{u}_1$ is a unit vector of segment s_{23} , and $\mathbf{u}_2$ is a unit perpendicular to segment s_{23} in a clockwise direction (figure 5).

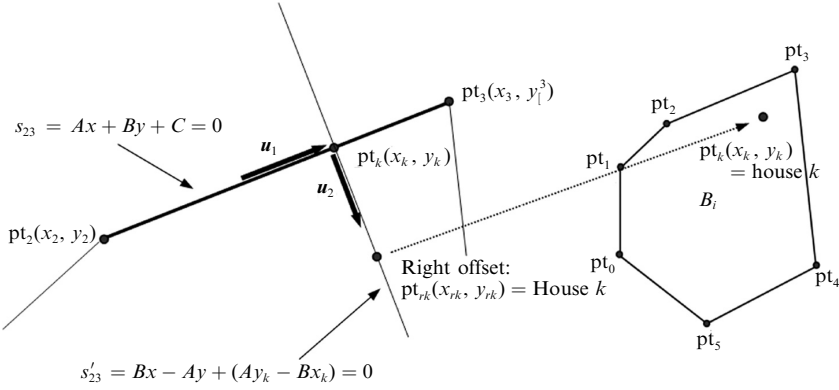


Figure 5. Right offset for 2D area-based address geocoding.

The cross product $\mathbf{u}_1 \times \mathbf{u}_2$ for two successive segment vectors is a vector perpendicular to the (xy) -plane, with a z -component equal to $-u_{1x}u_{2y} + u_{1y}u_{2x}$, which should be negative (otherwise, the point will be location outside the block). In order to avoid working in 3D, the ‘dot-perpendicular’ produce is used. This is a signed measure of the area of the parallelogram determined by two vectors

$$\begin{aligned} (u_{1x}, u_{1y}) \cdot^\perp (u_{2x}, u_{2y}) &= (u_{1x}, u_{1y}) \cdot (-u_{2y}, u_{2x}) = -u_{1x}u_{2y} + u_{1y}u_{2x} \\ &= z \text{ coordinate of } (u_{1x}, u_{1y}, 0) \times (u_{2x}, u_{2y}, 0) . \end{aligned}$$

In the final step, the (x, y) -coordinates of the new point pt_k (representing house k) with a right offset are defined by the mathematical functions described in the previous step (figure 4).

$$\mathbf{u}_1 \times \mathbf{u}_2 = -u_{1x}u_{2y} + u_{1y}u_{2x} ,$$

where unit vectors $\mathbf{u}_1 = (u_{1x}, u_{1y})$, and $\mathbf{u}_2 = (u_{2x}, u_{2y})$, and $\mathbf{u}_1 \times \mathbf{u}_2 < 0$ when the two vectors are in a clockwise direction, and $\times$ = dot-perpendicular product.

3.2 3D address geocoding method

Different geographic applications require different levels of accuracy regarding the location of geographic features. The geocoding method proposed in the previous section determines approximate locations of the given database, in much the same way as a street-address-matching method. One of the most precise methods is the matching of the input address to an object from the individual-point-address reference database (Ordnance Survey, 2004).

As seen in figure 6, each point in the building-address database represents a building that may be a single family house, an apartment complex, or a commercial building. While a single address is assigned to single family houses, multiple addresses are assigned to apartment complexes or commercial buildings. In order to assign addresses to apartment units or suites for buildings in urban areas, a 3D addressing system is used. The elements of a 3D address consist of two components, a subunit address and a building address. Subunit addresses, such as suite, room, or apartment numbers, describe the locations of a building’s interior rooms on each floor. The building address is represented by either a street addressing system, an area-based addressing system, or a hybrid addressing system. For example, the 3D address of an apartment unit in Charlotte, NC is 1205 Tryon Street #201, Charlotte, NC 28203, USA. The indoor address is #201 as an apartment number, and the rest of address represents the building address based on the street-addressing system.

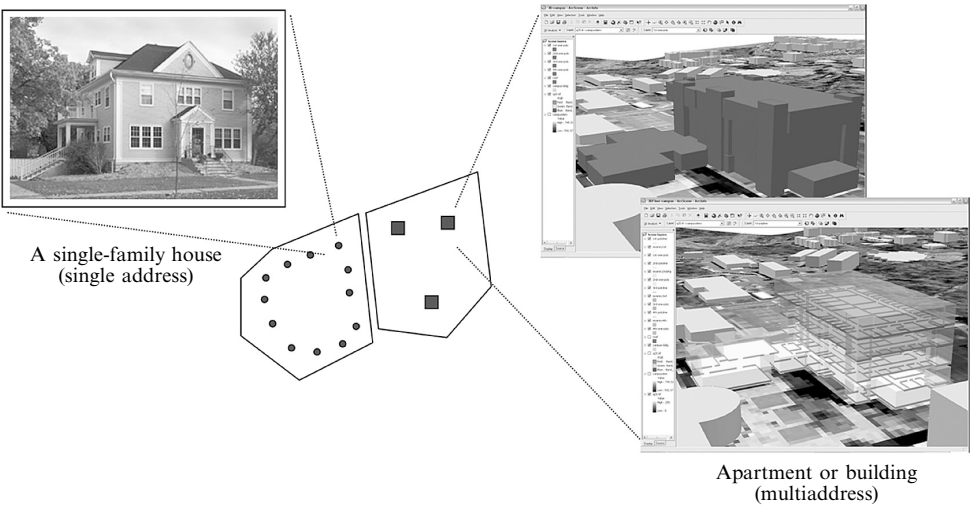


Figure 6. Hierarchical building point address.

For the 3D positioning, a 3D indoor geocoding technique (see Lee, 2004b) has been developed to identify the location of disaster events and conditions within a multilevel structure based on the descriptive location information for a real-time emergency-response decision-making system. Location information such as ‘there is a chemical explosion at 416 McEniry at the University of North Carolina at Charlotte’ obtained from 911 calls or emergency crews with wireless communication devices, can be transmitted back to the emergency-response system in real time. The 3D indoor geocoding method would then translate the descriptive location information into geographic positions within the building, the (x, y, z) -coordinates based on a given reference dataset. For implementing 3D positioning methods, microscale urban environments (such as the internal structure of a building) have been modeled as a network representation to serve as a frame of reference (reference data).

A useful 3D address geocoding method identifies and displays geographic coordinates from 3D postal addresses, based on a 3D indoor geocoding method (Lee, 2004b). In this system, the building address database contains the SingleAddressObject class, called a Point, and the MultiAddressObject class, called a Master_Point in a hierarchical structure (Mainguenaud, 1995). The MultiAddressObject represents a building address such as 1205 Tryon Street, Charlotte, NC 28203, USA. As seen in figure 6, the MultiAddressObject, a consolidated object of the internal structure of a building, includes attribute data on multiple apartment or suite addresses of the building. The following section describes how to model the 3D interior of a building to serve as reference data, and develops a 3D address geocoding method to position the geographic coordinates from the 3D postal addresses on the basis of the 3D indoor model.

3.2.1 3D indoor network model

In order to model the interior of a building in 3D to serve as reference data for geocoding an individual apartment or suite address in 3D, called a 3D indoor network model (3D-INM), a 3D geometric network model (3D-GNM) is utilized. This describes a topological model for representing connectivity relationships among 3D objects (Lee, 2001; 2004a). The model is based on a node-relation structure (Lee, 2001). The 3D-INM described here is based on the work of Lee (2001).

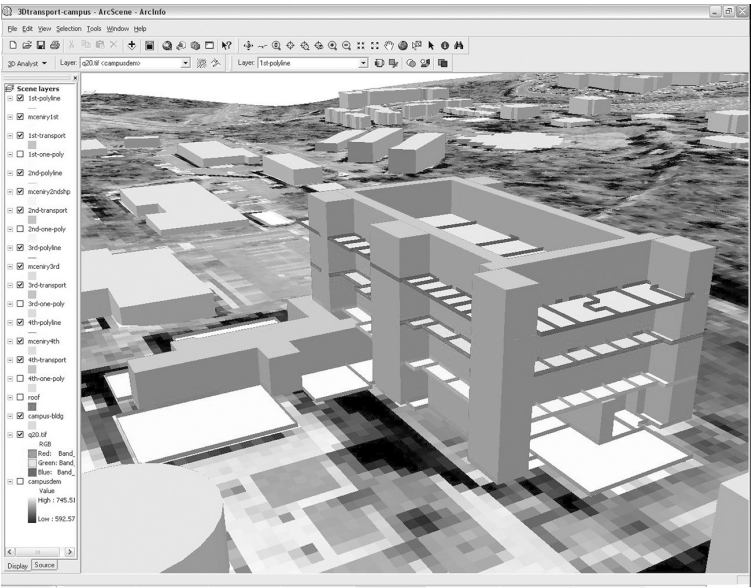
Instead of representing all internal structures of a building (including all rooms), transport routes such as corridors, stairways, and elevators within the building are modeled using a 3D-GNM. The connectivity relationships among 3D spatial units (transport routes within a building) are the combinations of the connectivity relations in the horizontal directions (on a floor) and the connectivity relations in the vertical direction (among floors). The connectivity relationships of 3D units in horizontal directions on a floor area are based on the connectivity relationships among polygons in 2D (for example, between corridors and stairways or elevators). The vertical connectivity relationships among the transport routes are defined by the locations of the stairway or elevators.

In order to model connectivity relationships among the transport routes on floor j , 3D Poincaré duality transformation (Lee, 2001) is utilized to define the dual graph of 3D objects. Through the duality transportation process, the connectivity relationships are modeled as a 3D network structure, the dual graph of 3D spatial units. A graph $G = [V(G), E(G)]$ consists of two sets, a finite set V of elements called edges, where $V(G) = \{v_1, \dots, v_n\}$, and $E(G) = \{(v_i, v_j)/v_i, v_j \in V\}$. Each edge is identified by a pair of nodes (v_1, v_2) , where $v_1, v_2 \in V$ (Mainguenaud, 1995).

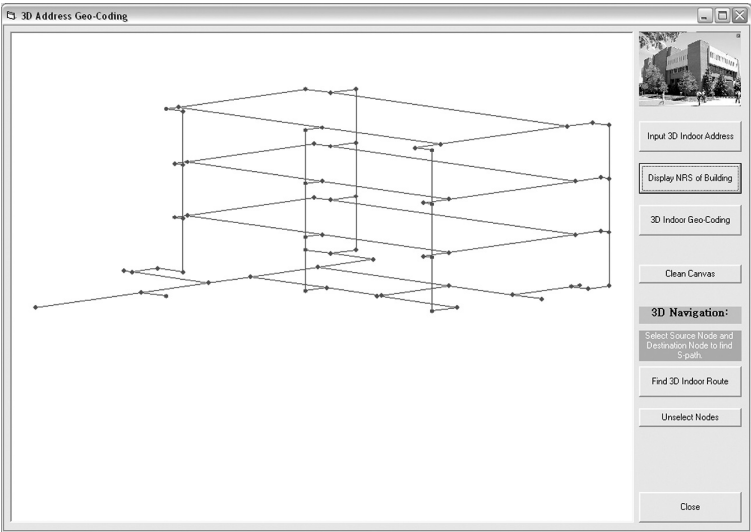
The dual graph does not represent the geometric properties (distance or size) of and among these units. In order to implement 3D address geocoding, the logical network model needs to be complemented by a 3D-GNM that represents accurately these geometric properties. To represent the transport routes using the 3D-GNM {a graph $Nh_j = [V(Nh_j), E(Nh_j)]$ }, the hallway needs to be transformed into linear features (Blum, 1967; Lee, 1982) based upon the straight medial axis transformation (Straight-MAT) algorithm {a graph $MAT_j = [V(MAT_j), E(MAT_j)]$ } (Aichholzer and Aurenhammer, 1988; Lee, 2004a). Next, each mode representing 3D spatial units (stairways or elevators) is projected and connected into the medial axis. The projection $p(q, e)$ of a point q onto an edge e is determined by the intersection of edge e and the edge perpendicular to edge e through the point q (Lee, 2004a).

The graph $Nh_j = [V(Nh_j), E(Nh_j)]$, representing the geometric network within floor j [figure 7(b)], is combined with the graph $Nv_n = [V(Nv_n)]$, which is a network representing the connectivity relations in vertical directions between floor $j-1$ and floor j . This combining of graphs executes a union operation to produce the graph $N_i = [V(N_i), E(N_i)]$ for a building i [figure 7(b)]. Figure 7(a) displays the transport routes within the study building in 3D view, and figure 7(b) depicts the extracted 3D-INM of the study area.

The 3D-INM, a graph $N_i = [V(N_i), E(N_i)]$, consists of a set of nodes V and a set of edges E . The primary classes of the model are NodeZ, EdgeZ, NetworkZ and MultiAddressObject, whose schema is shown in figure 8. the class NodeZ consists of an identifier, transport route types, and position data in 3D [(x, y, z)-coordinates]. The class EdgeZ consists of an identifier, a start node, an end node, and four subunit address ranges, which are the low and high suite numbers on the edge's left side and the low and high suite numbers on its right side. The subunit address ranges are used as reference data to identify the geographic positioning of a given 3D address using a 3D address geocoding method, which will be discussed in the following section. The class NetworkZ consists of an identifier and lists all nodes and all edges in a network. The MultiAddressObject in the database has an identifier, a building address, a building name (or building number, and a NetworkZ representing an abstract of the internal structure of the building and includes attribute data on multiple apartment or suite addresses of the building. Figure 9 describes the implementation of the 3D-INM in a relational database, and shows the complete logical model of the database structure.



(a)



(b)

Figure 7. Modeling the internal structure of a building: (a) transport routes within a building; (b) 3D geometric network model.

Each multiaddress object of the MultiAddressObject class in the database has a foreign key of the NetworkZ class.

3.2.2 3D address geocoding method

A 3D address consists of a building address (2D address) and a 3D indoor address. To determine the geographic locations of 3D address-based data, a two-step process can be executed—(1) to identify a MultiAddressObject (building) location based on Multiple_address dataset within the building address database, and (2) to determine the geographic location of the 3D indoor address within the MultiAddressObject (building).

```

class NodeZ {
    Int    NodeZ_ID;
    Double x, y, z;
    String TransportType
};

class NetworkZ {
    Int    NetworkZ_ID;
    NodeZ  ArrayNodeZ = new NodeZ[];
    EdgeZ  ArrayEdgeZ = new EdgeZ[];
};

class MultiAddressObject {
    Int    MPoint_ID;
    String Building_Address // Address for building
    String BuildingNum      // Building name
    Network Bldg_Network = new Network[];
};

class EdgeZ {
    Int    EdgeZ_ID;
    Float  Length;
    NodeZ  initial_node;
    NodeZ  end_node;
    Int    F-UA_l
    Int    T-UA_l
    Int    F-UA_r
    Int    T-UA_r
};

```

Figure 8. A 3D indoor network model (3D-INM).

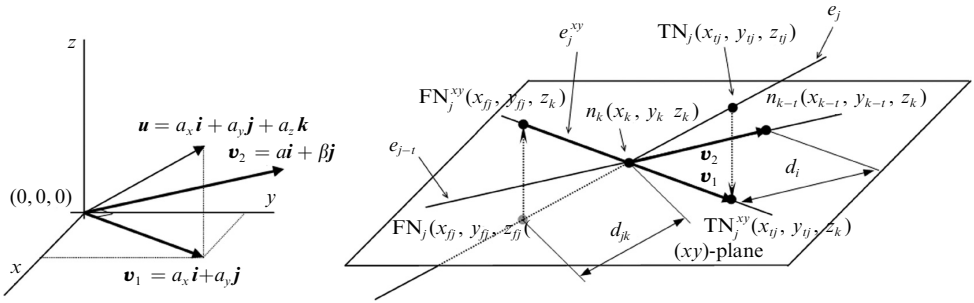


Figure 9. 3D address geocoding.

To geocode a 3D address of apartment unit 1205 Tryon Street #201, Charlotte, NC 29203, USA, the first step is to query (or search for) the apartment building from (in) the Multiple_address database on the basis of the building address, that is, 1205 Tryon Street, Charlotte, NC 28203, USA. The searching operation is based on the address-point-matching approach of the UK Ordnance Survey (2004). Using a point-georeferenced individual-address layer as the address reference database, the matching step establishes the corresponding relationships between an input address and all geographic information in the reference database. The individual point in the reference databases representing a single building has either a single address (2D address) or multiple addresses (3D addresses). When the retrieved building is a MultiAddressObject, the second step is necessary to geocode the 3D indoor address (such as #201) to determine the location of the apartment within the building, using the 3D address geocoding method.

The basic principle of the 3D address geocoding method uses a linear interpolation technique (US Census, 1992) used for the 2D geocoding method for area-based addressing systems. The internal structure of the building is modeled as a network, 3D-INM as seen in the previous section. An edge in the NetworkZ representing a hallway line segment contains apartment or suite addresses (unit numbers) as a range attribute, which refers to the low (F_UN_l) and high (T_UN_l) unit numbers on the edge's left side and the low (F_UN_r) and high (T_UN_r) unit numbers on its right side. Even though each unit in a building is uniquely named or labeled, the room or suite number is not always assigned as a number in sequence, unlike a street address. Before implementing the 3D geocoding, the room or suite numbers need to be standardized (using sequential numbering in clockwise order), by using translation tables in the parsing step of the geocoding process (Drummond, 1995).

In order to identify the location of the apartment or suite unit k within building i in the 3D-INM, $N_i = [V(N_i), E(N_i)]$, the 3D address geocoding method is executed in five steps. The procedure for geocoding a 3D postal address is summarized as follows:

Procedure. 3DAddressGeocoding (3DAddress $3D\text{-}Add_i$, MultiAddressObject $MPList$).
Step 1: Identify a MultiAddressObject (MP_i : a multiple address building) from the Multiple_address database based on the building address ($B\text{-}Add_i$): $3D\text{-}Add_i = \{Unit_k\} + \{B\text{-}Add_i\}$.

Step 2: Based on the target subunit number ($Unit_k$), query a specific edge (e_i) containing the subunit number $Unit_k$, by comparing it with the subunit number ranges of edges from the network N_i consolidating to the MP_i .

Step 3: Calculate a distance (d_{jk}) from From_Node (FN_i).

Step 4: Determine the location positioning (n_k) of $Unit_k$ [with (x, y, z) -coordinates] on the edge e_j .

Step 5: Apply a user-specified offset (d_i) to move the node location (representing a room) a certain distance away from the hallway line segment, and determine the final position (n_{k-l}) of $Unit_k$ (with x, y, z coordinates).

As seen in figure 9, the distance (d_{jk}) from From_Node (FN_j) to the node (n_k) of $Unit_k$ on the edge e_j is calculated on the basis of a linear interpolation. The distance d_{jk} calculated as the distance (d_j) between FN_j and TN_j multiplied by the proportion of house number ranges $[(Unit_k - F_UN_l)/(T_UN_l - F_UN_l)]$, if the $Unit_k$ is located at the left side of the hallway the edge e_j . The edge e_j connecting a node $FN_j(x_{fj}, y_{fj}, z_{fj})$ to a node $TN_j(z_{tj}, y_{tj}, z_{tj})$ is characterized by a vector: $\mathbf{v} = a_x\mathbf{j} + a_y\mathbf{j} + a_z\mathbf{k}$ in the direction of the edge, where $a_x = x_{tj} - x_{fj}$, $a_y = y_{tj} - y_{fj}$, and $a_z = z_{tj} - z_{fj}$. The (x, y, z) -coordinates of the node n_k (representing a subunit $Unit_k$) on the edge are calculated by the mathematical functions described below. $\mathbf{u}$ is a unit vector in this direction of the edge.

For a node $n_k(x_k, y_k, z_k)$,

$$x_k = x_{fj} + (d_{jk} \times b_x),$$

$$y_k = y_{fj} + (d_{jk} \times b_y),$$

$$z_k = z_{fj} + (d_{jk} \times b_z),$$

where $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\| = b_x\mathbf{i} + b_y\mathbf{j} + b_z\mathbf{k}$.

Because the geocoded position of the subunit $Unit_k(n_{k-l})$ represents the centroid of the subunit k , the node n_k defined in the previous step needs to be moved a user-specified offset distance (d_i) from edge e_j . Because the spatial units in question are residential or office units within a building, these spatial subunits are represented as 3D spatial objects with flat top and bottom faces that are parallel to the (x, y) -plane and perpendicular to the z -axis (Lee and Kwan, 2005). In terms of the nature of these spatial units, edge e_{j-l} connecting the node n_k to a geocoded node (n_{k-l}) is perpendicular to edge e_j and parallel to the (x, y) -plane. In order to identify the (x, y) -coordinates of the node (n_{k-l}), because $z_{k-l} = z_k$ the edge e_j needs to be projected to the (x, y) -plane passing through node n_k . As seen in figure 9, a vector in the direction of the edge e_j^{xy} connecting between a node $FN_j^{xy}(x_{fj}, y_{fj}, z_k)$ and a node $TN_j^{xy}(x_{tj}, y_{tj}, z_k)$ is $\mathbf{v}_1 = a_x\mathbf{i} + a_y\mathbf{j}$. Vector $\mathbf{v}_2$ is perpendicular to vector $\mathbf{v}_1$ on the

(x, y) -plane: $\mathbf{v}_2 = \alpha \mathbf{i} + \beta \mathbf{j}$. The (x, y) -coordinates of the node (n_{k-l}) can be calculated by the following two equations:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = (a_x, a_y) \cdot (\alpha, \beta) = (x_{ij} - x_{jj}) \times (x_{k-l} - x_k) + (y_{ij} - y_{jj}) \times (y_{k-l} - y_k) = 0,$$

$$d_j = [(x_{k-l} - x_k)^2 + (y_{k-l} - y_k)^2]^{1/2}.$$

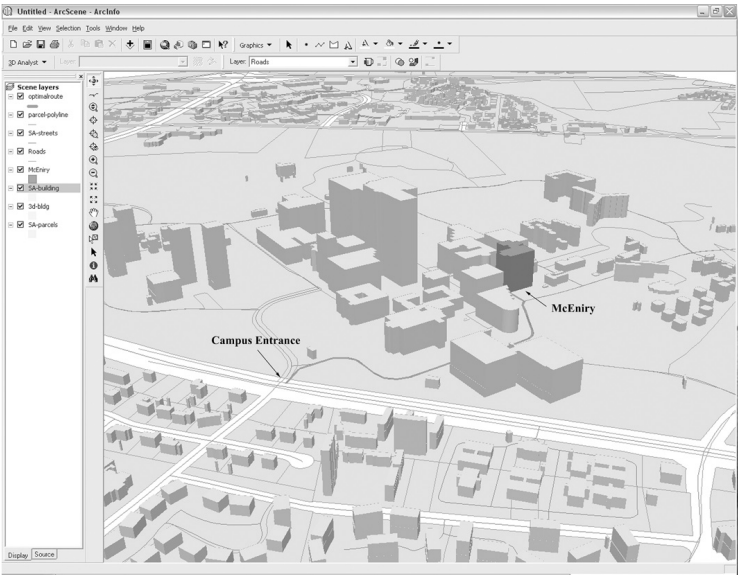
From these equations, two pairs of (x, y) -coordinates of the node (n_{k-l}) are solved: (x_{k-l}, y_{k-l}) and (x_{k-r}, y_{k-r}) . Suppose that the $\mathbf{v}_2$ is the vector in the direction of an edge between two nodes, (x_k, y_k) and (x_{k-l}, y_{k-l}) . If $\mathbf{v}_1 \times \mathbf{v}_2 > 0$ when the two vectors are counterclockwise, the node $n_{k-l}(x_{k-l}, y_{k-l}, z_{k-l})$ is the geocoded position of the subunit *Unit_k* with a left-offset from the hallway, while the $n_{k-r}(x_{k-r}, y_{k-r}, z_{k-r})$ is the geocoded position with a right-offset. If $\mathbf{v}_1 \times \mathbf{v}_2 < 0$, the opposite occurs, as described in the previous section. Depending on the signed measure of the area of the parallelogram determined by the two vectors, the 3D address geocoding method identifies the geographical coordinates of the 3D postal address.

4 Experimental implementation of a 3D address geocoding method

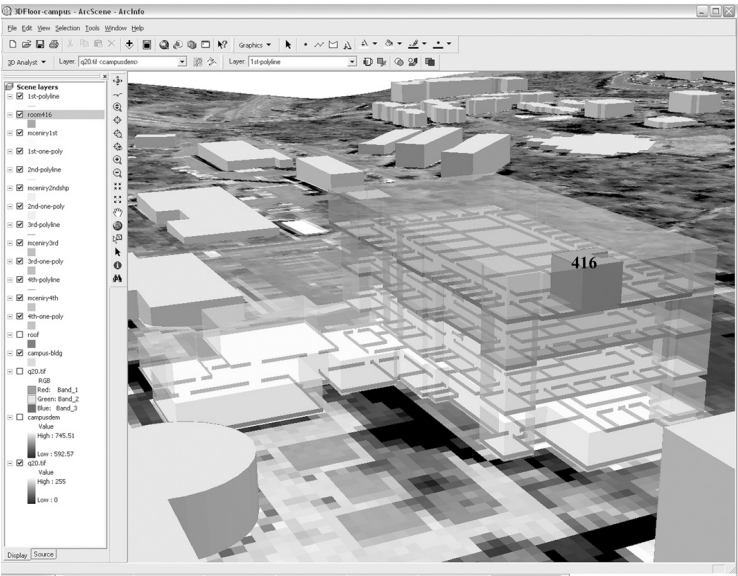
In order to demonstrate the benefits of using the 3D address-positioning method, this paper implements 3D indoor navigation to define optimal routes within microscale urban environments. The system has two components. The first component is a 3D visualization module for visualizing in 3D buildings and geographic features in urban areas, which utilizes ESRI ArcScene (ESRI, 2005) as the 3D viewers for displaying the urban features [figure 7(a)]. The second component is a 3D address geocoding module developed by Visual Basic programming language [figure 7(b)]. The dataset used is drawn from a comprehensive GIS database for Mecklenburg County (North Carolina, USA) and focuses on the area surrounding the McEnery Building at the University of North Carolina at Charlotte, USA. The 3D model of the building is based on the data model of an ESRI ArcView 3D shapefile, obtained with links to a GIS database including the building layout information. Figure 10(a) depicts the transport routes within the McEnery building in 3D Visualization Module, and figure 10(b) illustrates the extracted 3D-INM of the building. The model presents the connectivity relationship among transport routes inside the building.

Suppose that at the beginning of a semester, new students are looking for classroom 416 within the McEnery Building [labeled 416 in figure 10(a)]. In order to access the classroom effectively, students need to: (1) find where the building is located, (2) know where the room is located within the building, and (3) identify the optimal route from a source node (labeled South Entrance) to the destination node (classroom 416 on the 4th floor of the building) using 3D navigation.

The 3D address of the classroom within the building is 416 McEnery Building, 9201 University City Blvd, Charlotte, NC 28223. Using the building address, McEnery Building, 9201 University City Blvd, Charlotte, NC 28223, a matching step is implemented to find the most precise reference information (MultiAddressObject) from the reference address database (Building Address database). The matching process is an attribute query based on the building address. The outputs of the process are a Master_Point identifier (MPoint_ID) and the classroom number (416). Figure 10(a) shows the result of the attribute query based on the building address. The queried McEnery Building is located, and the figure also illustrates the result of an optimal-path analysis between an entrance of the university and the destination building. The network analyst module (ESRI, 2005), one of the ESRI ArcGIS extensions, was used to define the optimal route.



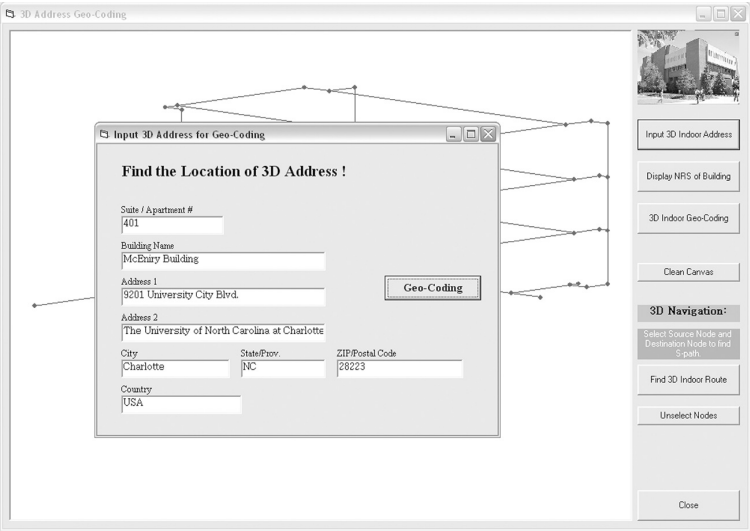
(a)



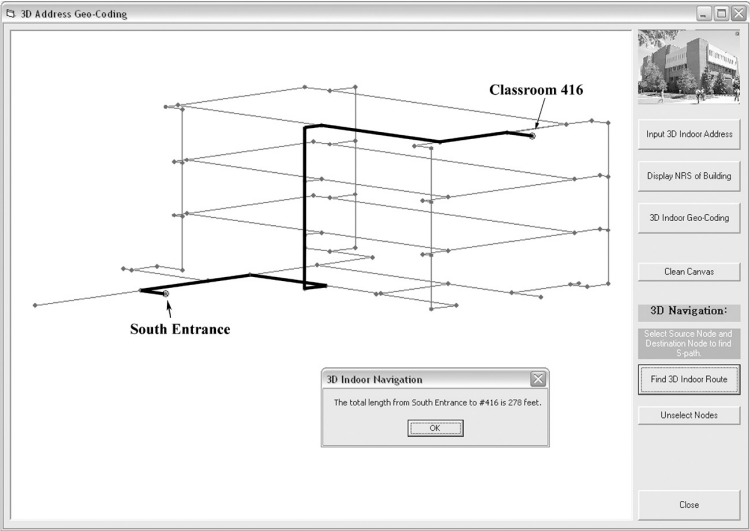
(b)

Figure 10. Study area displayed in 3D visualization module: (a) optimal route from campus entrance to the McEnery Building; (b) classroom 416 within the McEnery Building.

The second step is to identify the location of the classroom within the building. In the 3D data model of the building a detailed structural (CAD) plan would be available, so that the classroom could be queried and displayed in the 3D visualization module, as seen in figure 10(b). The position information of the classroom should be located in a 3D digital space to define the correlation with the 3D-INM, which represents the internal structures of the building as a network representation. A map-matching approach can be used to locate the classroom’s position relative to the linear features



(a)



(b)

Figure 11. 3D geocoding and optimal route searching: (a) address geocoding input dialog box; (b) geocoded classroom 416 and optimal route from the South Entrance.

of the 3D network (point-to-line matching) (Lee, 2004b; Zhou, 2005). However, although a 3D data model of the building is not available, we still need to identify the location information of the classroom within the building represented in the MultiAddressObject database by using the 3D address geocoding method.

To identify the location of a classroom (416), the users select the Input 3D Indoor Address button in the 3D address geocoding module of the system, in order to input to the system the 3D address of the building. The Input 3D Address for Geocoding dialog window is displayed in the Viewer area. As mentioned before, the McEnery Building is queried as a MultiAddressObject by the matching process based on the building address. Clicking on the Display NRS of Building button displays the

3D-INM of the building (McEniry). In order to geocode the classroom (416), the user selects the 3D Indoor Geo-coding button to execute the 3DAddressGeocoding (3DAddress 3D-Add, MultiAddressObject *MPList*) procedure described in the previous section which results in displaying the geocoded room location with the 3D network representing the 3D indoor model. The geocoded classroom is connected to the 3D-INM in order to define the correlation of the position with the 3D-INM, as seen in figure 11(b).

In order to demonstrate one of the benefits of using the 3D address-positioning method, a 3D indoor navigation is implemented to define optimal routes within the building. For this experimental implementation, Dijkstra's priority-first search (Dijkstra, 1959) for finding the shortest path in graphs, which was implemented in an NRS-System (Lee, 2004a), is applied. The figure illustrates the optimal route from the entrance to classroom 416. The optimal route was defined on the basis of the network's geometric distance as a travel cost, because the students would walk from the origin (South Entrance) to the destination (classroom 416). The students passing the entrance should turn right, walk along the hallway on the first floor of the building and turn right again to use elevators located at the west side of the building, and go up to the fourth floor. They should then turn right walk along the hallway, and turn left to the destination, classroom 416.

5 Conclusion

The currently used address geocoding methods are not universally applicable to developing a single portal geocoding middleware for worldwide Internet-based GIS applications, or have limitations in identifying features and incorporating 3D locational data from 3D addresses. In order to overcome these limitations in the address-matching methods, this paper proposed two additional address-matching methods, which are called an area-based address geocoding method and a 3D address geocoding method. The area-based address geocoding method is a 2D positioning method based on a 2D area-based address-matching technique and the 3D address geocoding method is based on a universal applicable 3D address geocoding technique. In order to demonstrate the benefits of using the 3D address-positioning method, this paper implements 3D indoor navigation to define optimal routes within microscale urban environments, as a useful by-product of the system.

While focusing on developing geocoding methods, this paper ignored several issues related to the implementation and deployment of the geocoding techniques. The first issue concerns the address-database scheme. In most countries (see, for example, Clallam County, WA, 2004) different addressing structures exist among rural-area addresses, urban-area addresses, single-family-house addresses, and 3D addresses including apartment or suite addresses. In order to maintain the address data consistently in databases, the geocoding system requires the development of a database schema for the address system. To resolve the problem, a government may wish to rationalize its address scheme by reassigning numerical addresses and modifying street names, or by transforming to other address systems (Eichelberger, 1993). However, the rationalization program can only be undertaken after evaluating the benefits of new addressing for citizens and businesses (Drummond, 1995). For most governments, address standardization will provide a better, less expensive and noncontroversial means of maintaining an accurate address database.

The next issue is related to the parsing step of geocoding methods. The house numbers in the area-based address systems and the subunit numbers in the 3D indoor addresses are not always assigned sequentially, unlike street addresses. This paper disregarded the need to design translation tables to transform unstructured address

data into structured and standardized address database records. In order to apply the proposed geocoding methods to real-world situations, the system should deal with issues relating to matching incomplete, out-of-date, inconsistent, or ambiguous addresses.

The third issue is to develop the actual addressing infrastructure with object classes such as point-georeferenced individual-building addresses, block data with house-number ranges, and MultiAddressObject database representing the internal structure of a building and including attribute data on multiple apartment or suite addresses of the building. When generating the 3D-INM, each node represents one subunit within the building no matter what size the unit is. Abstraction may be problematic for representing open-plan spaces, where networks do not naturally exist. When dealing with microscale geolocation within buildings based on the networks, geolocation away from corridors should be possible within the large open spaces. In order to extend the 3D-INM to resolve the problem, the nodes representing open-plan spaces need to be considered as a consolidated master_node, which is a subnetwork representing connectivity relationships among the compartmentalized zones of the open space.

The fourth is related to the measurement issues of potential errors. The proposed geocoding methods have the implicit assumption that all houses or buildings within a block or subunits along a hallway have the same length of frontage. Since the network distance is calculated from linear interpolations, unacceptable errors may occur when defining the geographical coordinates from the postal addresses. Due to assumptions in the indoor positioning technique, the proposed 3D geocoding method should be evaluated by applying any data-accuracy verification techniques prior to implementing the method. The other potential error in this technique is introduced when calculating optimal routes in a 3D network model. The proposed 3D network represents the geometric network properties (distances and directions) in horizontal routes, but not in vertical routes. The horizontal (x, y) spatial locations of vertical transport routes remains the same on each floor in the network model, however this can change significantly for stairways and escalators. In this implementation the optimal route was defined on the basis of the network's geometric distance as a travel cost, but in order to determine the most efficient route, a traffic time variable should be used to specify the weight of each edge in the 3D network (Kwan and Lee, 2005).

Next, in order to integrate with a 3D visualization system to allow the manipulation and exploration of georeferenced virtual environments, an additional data model is needed for geometric representation to visualize geographic entities in 3D viewers. This means that dual data models are required for implementing the 3D geocoding technique in a 3D GIS. In order to maintain the data consistency between two data models, 3D GISs require the development of database designs. Finally, the comprehensive GIS data to implement the 3D geocoding technique raise serious concerns about issues of data security, as the data can be misused by unauthorized people. Precautions that would prevent access to the data by such people should be considered before employing the system.

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References

- Aichholzer O, Aurenhammer F, 1998, "Straight skeletons for general polygonal figures in the plane", in *Proceedings of the Second Annual International Conference, COCOON '96, Hong Kong, Computing and Combinatorics, Lecture Notes in Computer Science 1090* Eds J-Y Cai, C K Wong (Springer, Berlin) pp 117–126
- Atkinson T D, 1999 *Merriam-Webster's Guide to International Business Communications* (Merriam-Webster, Springfield, MA)
- Blum H, 1967, "A transformation for extracting new description of shape", in *Symposium of Models for Perception of Speech and Visual Forms* (MIT Press, Cambridge, MA) pp 362–380
- Bullock B K, 1999, "Alternative sci.lang.japan frequently asked questions", <http://www.csse.monash.edu.au/~jwb/afaq/afaq.html>
- Clallam County, WA, 2004, "Clallam County's 911 addressing system", <http://www.clallam.net/RealEstate/html/addressing.htm>
- Corbett J P, 1979 *Topological Principles of Cartography* TP 48, US Census Bureau, Washington, DC
- Davis C, Fonseca F, Borges K, 2003, "A flexible addressing system for approximate geocoding", paper presented at *GeoInfo 2003* <http://www.geoinfo.info/geoinfo2003/papers/geoinfo2003-25.pdf>
- Dijkstra E W, 1959, "A note on two problems in connection with graphs" *Numerical Mathematics* **1** 269–271
- Drummond W J 1995, "Addressing matching: GIS technology for mapping human activity patterns" *APA Journal* (Spring) 240–251
- Eichelberger P, 1993, "The importance of address: the locus of GIS", in *Proceedings of the 1993 Conference of the Urban Regional Information Systems Association* **1** 212–217
- ESRI, 1998 *ESRI Shapefile Technical Description: ESRI White Paper* (ESRI Press, Redland, CA)
- ESRI, 2001 *ArcGIS User Guide* (ESRI Press, Redlands, CA)
- ESRI, 2005 *ArcGIS User Guide* (ESRI Press, Redlands, CA)
- Goodchild M, forthcoming, "Geographic information systems", in *International Encyclopedia of Human Geography* Eds R Kichin, N Thrift (Elsevier, Oxford)
- Knuth D E, 1973 *The Art of Computer Programming, Volume 3: Sorting and Searching* (Addison-Wesley, Reading, MA)
- Kwan M-P, 1998, "Space-time and integral measures of individual accessibility: a comparative analysis using a point-based framework" *Geographical Analysis* **30** 191–216
- Kwan M-P, 2000, "Interactive geovisualization of activity-travel patterns using three-dimensional geographical information systems: a methodological exploration with a large data set" *Transportation Research C* **8** 185–203
- Kwan M-P, Lee J, 2004, "Geovisualization of human activity patterns using 3D GIS: a time-geographic approach", in *Spatially Integrated Social Science: Examples in Best Practice* Eds M Goodchild, D Janelle (Oxford University Press, Oxford) pp 48–66
- Kwan M-P, Lee J, 2005, "Emergency response after 9/11: the potential of real-time 3D GIS for quick emergency response in micro-spatial environments" *Computers, Environment and Urban Systems* **29** 93–113
- Lee D T, 1982, "Medial axis transformation of a planar shape" *IEEE Transactions on Pattern Analysis and Machine Intelligence* PAMI-4 (4) 363–369
- Lee J, 2001 *A 3D Data Model for Representing Topological Relationships Between Spatial Entities in Built-Environments* unpublished PhD Dissertation, Department of Geography, The Ohio State University, Columbus, OH
- Lee J, 2004a, "A spatial access oriented implementation of a topological data model for 3D urban entities" *GeoInformatica* **8** 235–262
- Lee J, 2004b, "3-D GIS for geo-coding human activity in micro-scale urban environments", in *Geographic Information Sciences: Springer's Lecture Notes in Computer Science Computers (LNCS 3234)* Eds M Egenhofer, C Freksa, H Miller (Springer, New York) pp 162–178
- Lee J, Kwan M-P, 2005, "A combinatorial data model for representing topological relations among 3D geographic features in micro-spatial environments" *International Journal of Geographical Information Science* **19** 1039–1056
- Mainguenaud M, 1995, "Modeling the network component of geographical information systems" *International Journal of Geographical Information Systems* **9** 575–593
- Miller H J, 1999, "Measuring space-time accessibility benefits within transportation networks: basic theory and computational procedures" *Geographical Analysis* **31** 187–212
- MOGAHA, 2003 *Report on Korea Address System Planning* Ministry of Government Administration and Home Affairs, South Korea
- Ordnance Survey, 2004, "OSMasterMap address layer user guide", <http://ordnancesurvey.co.uk>

-
- Ordnance Survey, 2006, "Ordnance survey produce names and trademarks", <http://www.ordnancesurvey.co.uk/oswebsite/business/copyright/docs/productnamesandtrademarks.pdf>
- Pu S, Zlatanov S, 2005, "Evacuation route calculation of inner buildings", in *Geo-information for Disaster Management* Eds P J M van Oosterom, S Zlatanov, E Fendel (Springer, Heidelberg) pp 1143 – 1161
- UPU, 2004, "Postal addressing systems in member countries", Universal Postal Union, http://www.upu.int/post_code/en/postal_addressing_systems_member_countries.shtml
- US Census, 1992 *TIGER/Line Census Files, 1992: Technical Documentation* Department of Commerce, Washington, DC
- Walls M D, 2003, "Is consistency in address assignment still needed?", in *Proceedings of URISA 5th Annual GIS in Addressing Conference: Street Smart and Address Savvy Province, RI*, <http://www.worldbank.org/urban/upgrading/docs/urisa/Proceedings.pdf>
- Weber J, Kwan M-P, 2002, "Bringing time back in: a study on the influence of travel time variations and facility opening hours on individual accessibility" *The Professional Geographer* **54** 226 – 240
- Winter S, Iglesias G, Zlatanov S, Kuhn W, 2005, "GI for the public: the terrorist attack in London" *Geoinformatics Magazine* <http://www.gdmc.nl/zlatanov/publications.htm>
- Zhou J, 2005, "A three-step general map matching method in GIS environment: travel/transportation study perspective" *UCGIS Summer Assembly 2005* <http://www.ucgis.org/summer2005/studentpapers/zhoub.pdf>

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